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[23TCSDE11]

THIRD SEMESTER B.Tech EXAMINATION, NOVEMBER-2024

NETWORK DEFENCE

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	Define application-level attacks and list common types.	L1	CO1	**02**
	b.	How would you use an IDS to monitor activities at the network perimeter?	L2	CO2	**02**
	c.	What are the benefits of network virtualization in a data center environment?	L1	CO3	**02**
	d.	What is the purpose of network traffic monitoring in cybersecurity?	L1	CO4	**02**
	e.	How risk anticipation can improve overall risk management outcomes?	L1	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Analyze how social engineering attacks exploit human trust to achieve their objectives.	L4	CO1	**05**
		OR			
	b.	Describe how phishing attacks compromise email security.	L1	CO1	**05**
3.	**a.**	What is mobile application security, and What strategies can organizations employ to mitigate the risks associated with malicious or vulnerable mobile apps	L2	CO2	**05**
		OR			
	b.	Explain the primary components of an effective endpoint security solution for windows systems.	L1	CO2	**05**
4.	**a.**	Explain how network virtualization enhances user productivity.	L2	CO3	**05**
		OR			
	b.	Describe the role and attacks of Virtual LANs (VLANs) in network virtualization security.	L2	CO3	**05**
5.	**a.**	Evaluate the effectiveness of Log Monitoring and Analysis on Windows Systems and Linux System.	L3	CO4	**05**
		OR			
	b.	How would you implement a network performance and bandwidth monitoring solution in a small office network?	L2	CO4	**05**
6.	**a.**	Explain the significance of understanding attack surfaces in an organization's security strategy.	L2	CO5	**05**
		OR			
	b.	illustrate an understanding of different types of disasters and their potential impact on business operations.	L2	CO5	**05**
7.	**a.**	Apply the concept of virtualization in cloud computing and develop a best practice for securing data in a cloud environment.	L3	CO6	**05**
		OR			

	b.	Evaluate the impact of risk management phase on achieving organizational goals.	L5	CO6	**05**
		PART-C	**6 X 10=60**		
8.	a.	Illustrate the various types of Network level Attacks with defence Strategies in details.	L2	CO1	**10**
		OR			
	b.	Assess the strengths and weaknesses of your organization's current administrative or and technical security measures.	L4	CO1	**10**
9.	a.	Analyze the impact of endpoint firewalls on system performance and user experience.	L4	CO2	**10**
		OR			
	b.	Describe the main objectives of data security, including data confidentiality, integrity, and availability. Why are these objectives collectively known as the CIA triad?	L1	CO2	**10**
10.	a.	Explain the different types of wireless network authentication methods.	L2	CO3	**10**
		OR			
	b.	Classify wireless networks based on their coverage area, and use and explain in detail.	L1	CO3	**10**
11.	a.	Assess the impact of a Network Traffic Monitoring and Analysis strategy in meeting the organization's performance goals.	L3	CO4	**10**
		OR			
	b.	How would you implement a centralized log monitoring solution for a multi-site organization?	L2	CO4	**10**
12.	a.	Illustrate the significance of various phase in Incident Response process. Why is each phase essential for good incident management?	L2	CO5	**10**
		OR			
	b.	illustrate the significance of threat intelligence layer to enhance the organization's cybersecurity posture.	L2	CO5	**10**
13.	a.	Assess the effectiveness of encryption as a security measure in protecting sensitive information during transmission over the internet.	L4	CO6	**10**
	b.	Design a comprehensive application security policy for a software development organization that addresses user access, data protection, and incident response.	L6	CO6	**10**